

STRUCTURED PROMPTING LAB

Exercise 1: Role Prompting (25 min) -- Level 1

Compare all three outputs. Answer:

1. Which explanation would help YOU the most right now? Why?
 2. What changed between outputs? (Length? Tone? Detail? Examples?)
 3. Which role produced the most useful code example?
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1. Of the three Prompt Roles, the one that helped me the most is the third prompt. The technical writer role: explained variables the best because of not only the role, but also the constraints, the syntax for declaration and the code example as well as the common mistakes.
 2. The length changed. The examples changed. The detail changed and the tone changed. The examples for the teacher role were not helpful and having no role was too long of an explanation.
 3. The third prompt-Technical role, gave the best output example.

Exercise 2: Constraint Prompting (25 min) -- Level 1

Constraints force AI to work within boundaries. This is how you get precise output.

1. Count the lines. Did it stay under 50?
 2. Open it in Replit. Does it look right?
 3. Check on mobile (Replit preview on phone, or resize browser window). Is it readable?
 4. **Modify:** Change the color scheme to your own preference. Change the "My Goal" section to be about YOUR actual goal.
-
1. Yes, the lines stayed under 50 lines.
 2. It looks okay. The prompt didn't have a lot constraints, details, and left a lot to the imagination.
 3. Yes, it worked on my iPhone 17

Exercise 2: Constraint Prompting (25 min) -- Level 1

After AI generates the code:

1. Count the lines. Did it stay under 50? Yes, it stayed under 50 lines
2. Open it in Replit. Does it look right? It looks very basic
3. Check on mobile (Replit preview on phone, or resize browser window). Is it readable? It worked on my iPhone 17
4. **Modify:** Change the color scheme to your own preference. Change the "My Goal" section to be about YOUR actual goal. Modifications made.

Exercise 3: System Instruction Practice (20 min) -- Level 2

- A. **Scenario A:** You want AI to help you debug code. It should ask questions before giving answers, never rewrite your entire code, and explain errors in simple language.
 - a. Task Completed. It found the syntax errors and code errors.
- B. **Scenario B:** You want AI to help you brainstorm app ideas. It should give you 5 ideas at a time, explain who would use each one, and rate the difficulty from 1-5.
 - a. **RATE 1:** AI gave me my own past project information. For example: AI Neighborhood Resource Matcher. People post needs/offers(for the general public)
 - b. **RATE 5:** AI gave me my own past employment to tailor it to an app. AI Policy / Contract Explainer (YOUR EDGE). Leverage your insurance background. upload policy or contract and let AI simplify the information **[TERRIBLE IDEA]**
 - c. **RATE 1:** AI That Transforms Input → Output. **Output:** polished alerts (email, SMS, notifications). Eg. Water outage in Dallas.
 - d. **RATE 5:** AI Scenario Game Generator (refined version). **[I came up with this idea and asked it earlier how to create something like it]**
 - e. **RATE 2:** Smart Dog Walking App (upgrade your project). Based on the dog walking business page that I had it generate code for.

Scenario C: You want AI to quiz you on what you learned this week. It should ask one question at a time, wait for your answer, then tell you if you're right with an explanation.

It gave me this problem:

```
const students = [  
  { name: "Anna", score: 92 },  
  { name: "Ben", score: 76 }  
];  
  
for (let i = 0; i < students.length; i++) {  
  const student = students[i];  
  
  if (student.score > 80) {  
    console.log(student.name + " passed");  
  }  
}
```

PEER ACTIVITY: CONTEXT COMPARISON

Pairs. Each pair gets the same task.

Exercise 3: Build with Judgment (*remaining time*) -- Level 3

Round 1 -- No context (5 min) Completed.

Round 2 -- Rich context (10 min): Now rewrite the prompt with as much context as you can imagine:

<https://jdbostonbu-ops.github.io/Dog-Walking-Business/>

- Look at both outputs side by side
- Which one would you actually use? Why? I would use the 2nd prompt. I made a GitHub Repository for it.
- What specific details from the rich prompt showed up in the output? Color, Hero Section, warm colors, specific sections, mobile friendly. Landing page, specific quotes, prices, etc.

APPLIED JUDGMENT LAB

Exercise 1: Accept, Reject, or Modify? (40 min) -- Level 1

Scenario A: You asked AI to write a professional email to a potential employer. AI wrote:

Questions:

- Would you send this email? Why or why not? I wouldn't because it doesn't describe my true skills and training.
- What's wrong with it? (Hint: does this describe YOU right now? Is any of this true yet?) It says I have expertise in areas that I do not have.
- How would you modify it to be honest AND compelling? I would tell the truth, and explain my strengths and how they can be applied to software development. I would not list the languages as if I was an expert. I say that I am learning these languages and state specific dates emphasizing how long I have been learning for.

Scenario B: You asked AI to write a JavaScript function that checks if a number is prime. AI wrote:

Honestly, the email is almost identical to the ones I receive from employers. It isn't easy writing an email that rejects a candidate. It highlighted strengths before turning him down.

Questions:

- Does this code work? (Yes -- it's correct and optimized)
- Do you understand every line? (Probably not -- the optimization in the loop is advanced) I understand the logic, but I would test it because I am not good at prime numbers. I would look up prime numbers first.
- Should you accept this? Or ask for a simpler version you can understand? I would then test to see if the prime numbers are generated from the code by running the code and comparing it to prime numbers.

Scenario C: You asked AI to suggest names for your new app that helps people find local food trucks. AI suggested:

Questions:

- Are any of these good? Which one would you pick? Yes, they were good choices.
- Do any of them already exist? (How would you check?) I would google them in the state data base. I would ask AI to search the database to see if they exist.
- What would YOU name it? Is your name better than AI's suggestions? Honestly, I am not good at naming businesses. No, AI's suggestions were better than mine.

Exercise 2: The Ethics Scenarios (30 min) -- Level 2

Scenario 1: The Resume.

- What went wrong? The resume and cover letter was not reviewed.
- Where should you have drawn the line? You draw the line on the first draft.
- How would you use AI for job applications without crossing that line? I would use my resume that I created. Then, ask AI to review the job posting to see if my skills and experience match the position. I would ask AI to write the cover letter only if it found I was a good match for the position.

Scenario 2: The Client Project.

- What should you have done differently during the build? Never accept something that you didn't review, read and test yourself. You are accountable and own the product that AI creates based on your prompt.
- What's the minimum level of understanding you need to deliver responsibly? You need to be able to read and understand and test what AI generates.

Scenario 3: The Bias Check.

- What assumptions did AI make about gender, appearance, or background? AI made an assumption that they were gender neutral, but it is my understanding that the software engineer field is dominated by men. Nurses are also dominated by women. It made these fields appear like they were good fields and were made up of intelligent, organized individuals
- Where do these assumptions come from? The assumptions come from generating all search in the internet.
- How does this affect people who don't match the stereotype? It makes a huge impact on people because if you asked it about tele-marketers or insurance agents, it would paint a negative picture based on bias information during its search.

Exercise 3: Build with Judgment (*remaining time*) -- Level 3

About Me Prompting:

<https://jdbostonbu-ops.github.io/About-Me/>

AI Output	My Decision	Why
Generated a bio that says "software engineering in training"	Rejected -- changed to Learning to build with AI. With over 15 years of experience in data analysis and pattern recognition, I'm transitioning into AI Software Development, blending my background with modern tools to create user-focused, intelligent solutions.	I'm not an expert yet. Honesty matters.
Suggested a color scheme I don't like	Modified --Changed everything. Changed minimalist to modern. Asked AI to use different font sizes, hover, animation, shadow box, added resume file, instructed to not make the landing page like a resume but utilize the resume for information gathering.	My page, my taste.
Wrote a "Skills" section listing React, Node, SQL	Rejected -- too early. Changed to "Currently Learning"	I haven't learned those yet. Will add them when I do. Added repositories
Generated clean HTML structure	Rejected—Changed html to three files. Issues with GitHub push	The structure works and I understand every tag.

Peer Activity: Explain AI to a Non-Technical Person

What is AI and how does it work?

Your partner scores you:

- Did I understand the explanation? (yes/no) Yes
- Was there any jargon I didn't follow? (what?) No

- What was the strongest part? That AI is a machine that uses information that is readily available on the internet. It doesn't use reliable information and must be used with caution.